IaC Provisioning for Real Estate System

Course Name:

DevOps

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2026

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Introduction

This document presents the High-Level Design (HLD) of the IaC Provisioning for Real Estate System. The primary goal of this project is to automate the provisioning, configuration, and deployment of infrastructure required to host a Real Estate application using DevOps practices.
Traditional infrastructure setup involves manual configuration, which is time-consuming and error-prone. This project leverages Infrastructure as Code (IaC) tools such as Terraform and Ansible to automate infrastructure creation and configuration. Additionally, containerization using Docker and CI/CD automation using Jenkins or GitHub Actions ensures seamless deployment.
The system supports both cloud-based deployment (AWS) and local containerized environments, making it flexible for development and production usage.

1. Scope of the Document

This document includes:

- Design of automated infrastructure provisioning
- Deployment strategy using Docker containers
- Integration of CI/CD pipelines
- Cloud infrastructure setup using AWS
- Configuration management using Ansible
- Security, scalability, and performance considerations
- High-level workflows and architecture

This document does not include:

- Detailed source code implementation
- UI/UX design of the Real Estate application
- Business logic of the application

2. Intended Audience

This document is intended for:

- DevOps Engineers – to understand automation workflows and infrastructure design
- Developers – to understand deployment pipelines and environment setup
- System Administrators – to manage deployed infrastructure
- Project Evaluators – to assess system design and architecture

- Cloud Engineers – to understand provisioning strategies

Basic familiarity with Docker, AWS, and CI/CD is assumed.

3. System Overview

The system automates the deployment lifecycle of a Real Estate application. It integrates multiple DevOps tools to ensure that infrastructure provisioning, configuration, and deployment are executed automatically.

Key functionalities include:

- Provisioning cloud infrastructure (EC2, VPC, IAM, S3)
- Configuring servers using Ansible
- Containerizing applications using Docker
- Automating deployments using CI/CD pipelines
- Monitoring deployed services

Key benefits:

- Eliminates manual errors
- Ensures consistency across environments
- Enables rapid deployment
- Supports scalability and high availability

System Design

1. Application Design

The system follows a layered and modular architecture, where each layer performs a specific role independently but integrates seamlessly.

Layers Explained:

Infrastructure Layer (Terraform):
Responsible for provisioning cloud resources like EC2 instances, networking components, and storage.

Configuration Layer (Ansible):
Installs required software, dependencies, and configures the servers after provisioning.

Container Layer (Docker):
Packages the application into containers ensuring portability and consistency.

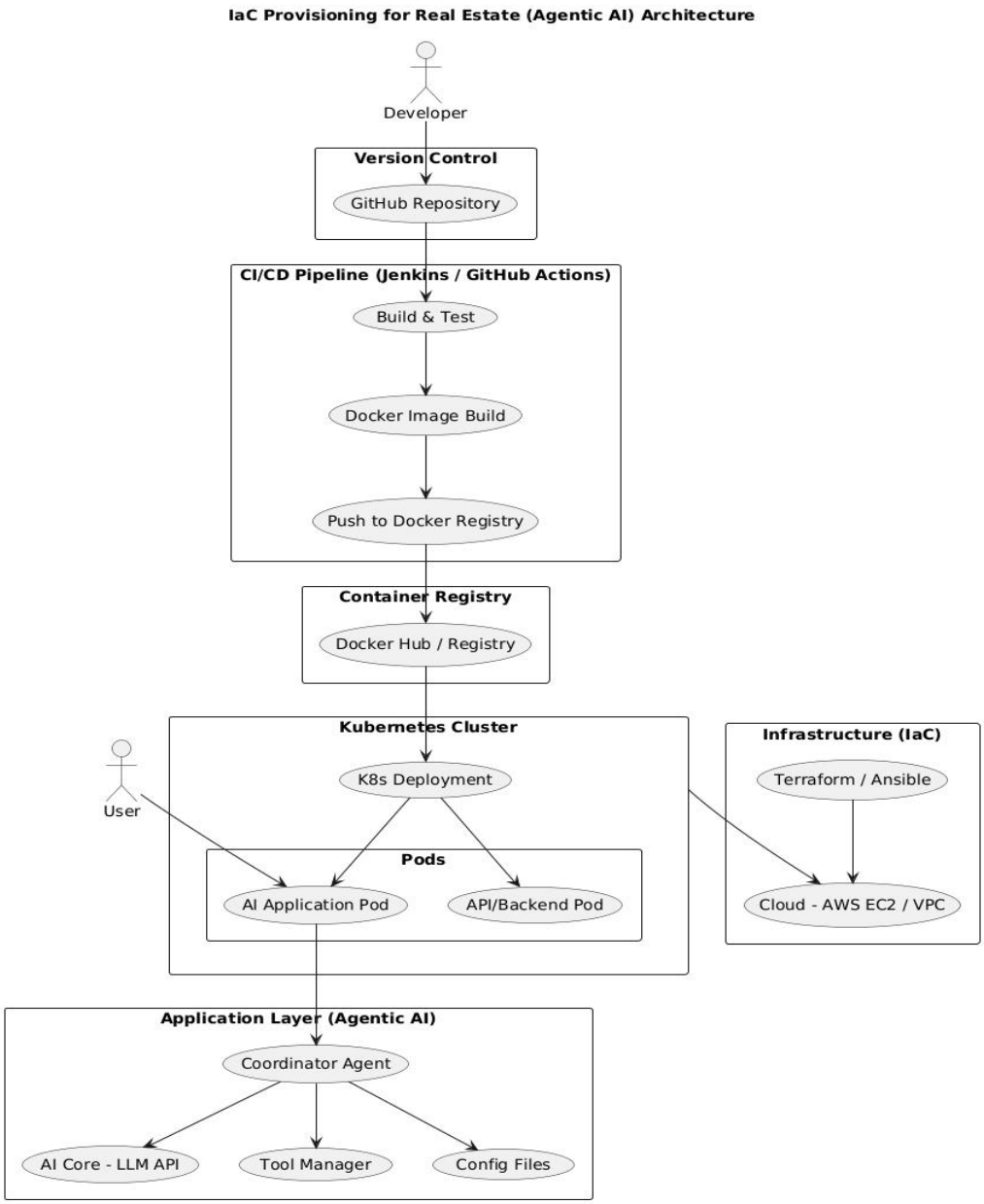
CI/CD Layer (Jenkins/GitHub Actions):
Automates build, test, and deployment processes.

Version Control Layer (Git):
Maintains source code, infrastructure code, and configuration scripts.

Execution Layer (AWS/Local):
The environment where the application is deployed and executed.

Design Approach:

- Modular Terraform modules for reusability
- Ansible roles for structured configuration
- Stateless container-based deployments
- Separation of concerns between infrastructure and application



2. Process Flow

The workflow ensures complete automation from code commit to deployment.

Detailed Flow:

- Developer pushes code to Git repository
- CI/CD pipeline is triggered automatically
- Pipeline validates code and builds Docker image
- Terraform initializes and provisions infrastructure
- Infrastructure (EC2, networking) is created on AWS
- Ansible connects to instances via SSH
- Required dependencies (Docker, runtime) are installed
- Docker containers are deployed on provisioned instances
- Application becomes accessible to end users
- Logs and monitoring tools track system health

3. Information Flow

The system ensures smooth data and control flow between components:

- Source Code Flow: Developer → Git → CI/CD
- Infrastructure Instructions: Terraform → AWS APIs
- Configuration Flow: Ansible → EC2 Instances
- Application Flow: Docker → Running Containers
- Monitoring Flow: Logs → Monitoring tools

This structured flow ensures traceability and automation.

4. Component Design

Git Repository

- Stores application code, Terraform scripts, and Ansible playbooks
- Enables version control and collaboration

Terraform

- Defines infrastructure in code (HCL)
- Creates AWS resources like EC2, VPC, S3

Ansible

- Automates configuration and software installation
- Ensures idempotent execution

Docker

- Packages application and dependencies
- Ensures environment consistency

CI/CD Tool (Jenkins/GitHub Actions)

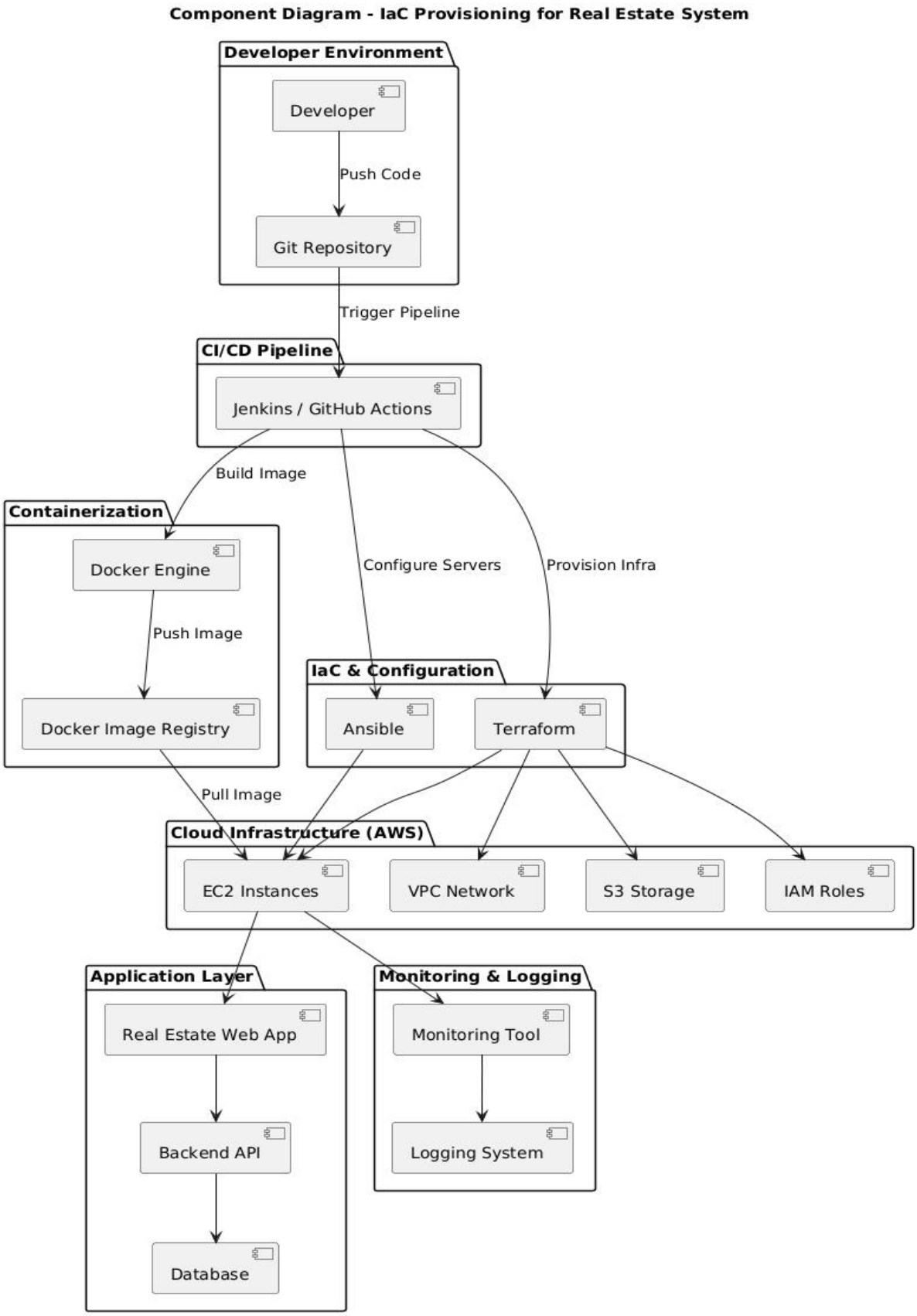
- Automates build and deployment
- Integrates all components

AWS Cloud

- Hosts infrastructure
- Provides scalability and reliability

Monitoring & Logging

- Tracks performance and failures
- Enables alerting mechanisms


Component Diagram - IaC Provisioning for Real Estate System
Developer Environment
Developer
Push Code
Git Repository
Trigger Pipeline
CI/CD Pipeline
Jenkins / GitHub Actions
Build Image
Configure Servers
Provision Infra
Containerization
Docker Engine
Push Image
Docker Image Registry
IaC & Configuration
Ansible
Terraform
Pull Image
Cloud Infrastructure (AWS)
EC2 Instances
VPC Network
S3 Storage
IAM Roles
Application Layer
Real Estate Web App
Backend API
Database
Monitoring & Logging
Monitoring Tool
Logging System

5. **Key Design Considerations**

- Scalability: Ability to handle increased user load using auto-scaling
- Security: Use of IAM roles, encrypted storage, secure communication
- High Availability: Use of multiple availability zones
- Modularity: Independent and reusable components
- Fault Tolerance: System recovery mechanisms
- Cost Efficiency: Use of optimized cloud resources
- Automation: Minimal manual intervention
- Consistency: Same setup across all environments

6. **API Catalogue**

S.No	API Name	Purpose	Method	Description
1	AWS EC2 API	Instance Creation	POST	Launch virtual machines
2	AWS VPC API	Network Setup	POST	Configure networking
3	AWS S3 API	Storage	PUT/GET	Store data and state files
4	Docker API	Container Mgmt	REST	Manage containers
5	Terraform CLI	Infra Execution	CLI	Apply infrastructure code
6	Ansible SSH	Config Mgmt	SSH	Configure instances
7	Jenkins API	Pipeline Trigger	POST	Automate workflows

Data Design

1. Data Model

The system handles both infrastructure and application-level data.

Types of Data:

- Terraform state files
- Configuration files (YAML, JSON, HCL)
- Application data (property listings, users)
- Logs and monitoring data

Entities:

- Infrastructure resources

- Deployment configurations
- User data
- Logs

2. Data Access Mechanism

- Secure SSH-based communication
- IAM-based authentication
- Role-based access control
- Encrypted storage using AWS S3
- API-based communication

3. Data Retention Policies

- Terraform state stored with versioning in S3
- Logs retained for audit (30–90 days)
- Backup strategies (daily/weekly)
- Sensitive data encrypted

3. Data Migration

- Automated migration scripts
- Backup and restore mechanisms
- Version-controlled schema updates
- Zero downtime deployments

Interfaces

The system supports multiple interfaces:

- CLI (Terraform, Ansible)
- Cloud Console (AWS Management Console)

- Web Interface (Real Estate Application)
- REST APIs
- Monitoring Dashboards

State and Session Management

- Terraform remote backend using S3 with state locking
- Stateless application containers
- Session management using database or Redis
- IAM-based session handling

Caching

- Redis for application-level caching
- Docker layer caching for faster builds
- Terraform plugin caching
- CDN caching for faster content delivery

Non-Functional Requirements

1. **Security Aspects**
    - IAM role-based access control
    - Encrypted data storage (S3 encryption)
    - HTTPS secure communication
    - SSH key-based authentication
    - Secrets management (API keys, credentials)
    - Firewall and security groups

2. **Performance Aspects**
    - Auto-scaling groups for handling load

- Load balancers for traffic distribution
- Optimized Docker images
- Parallel infrastructure provisioning
- Real-time monitoring and alerts

References

- HashiCorp, Terraform Documentation. Available: https://developer.hashicorp.com/terraform/docs
- Red Hat, Ansible Documentation. Available: https://docs.ansible.com/
- Docker, Inc., Docker Documentation. Available: https://docs.docker.com/
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